GOKUL D

Embedded Systems Engineer | AI & IoT Developer

Coimbatore, Tamil Nadu, India | gokuldevadasan154@gmail.com | +91 9400039456

GitHub: github.com/Gokuldevadasan LinkedIn: linkedin.com/in/gokul-devadasan-7134a2328

PROFESSIONAL SUMMARY

Third-year Bachelor of Engineering (B.E.) student in Electronics and Communication Engineering with hands-on experience in embedded systems and firmware development using Embedded C across ESP32, STM32, and Arduino platforms. Skilled in building IoT and cloud-connected systems, with additional exposure to artificial intelligence, computer vision, and robotics through academic projects. Practical exposure to industrial and marine automation gained through internships, along with a published research paper on automation, machine learning, and embedded systems. Seeking an Embedded Systems, Firmware, IoT, or AI and Robotics internship to apply these skills to real-world engineering solutions.

TECHNICAL SKILLS

1. **Programming Languages:** Embedded C, C++, Python, Dart, Java, JavaScript
2. **Embedded Systems & Microcontrollers:** ESP32, STM32, Arduino
3. **IoT & Cloud:** Firebase, REST API, Flask, JSON
4. **Artificial Intelligence & Computer Vision:** OpenCV, OpenRouter API, LLM Integration, Computer Vision
5. **Mobile Development:** Flutter, Android Studio, Shared Preferences
6. **Electronics & Hardware:** Sensor Interfacing, PCB Design, Circuit Debugging, Control Panel Wiring, Marine Electronics
7. **Development Tools:** VS Code, Git, GitHub, Arduino IDE
8. **Operating Systems:** Windows, Linux, Ubuntu

EDUCATION

Bachelor of Engineering - Electronics and Communication Engineering | *Adithya Institute of Technology, Coimbatore (Anna University)*
2024 - 2028 | Third Year | CGPA: 8.38

INTERNSHIP EXPERIENCE

Assistant to Senior Service Engineer - INS Vikrant | *Navicom Technology International Pvt. Ltd.*
May 25, 2026 - June 05, 2026 | Marine Electronics, Navigation & Communication Systems

- Assisted the Senior Service Engineer with maintenance and routine servicing of navigation and communication equipment.
- Gained practical exposure to marine electronics systems associated with the Indian Naval Ship INS Vikrant at Karwar Naval Base.
- Supported technical activities related to communication equipment maintenance in a marine engineering environment.

Industrial Trainee - Marine Automation | *SMEC Automation Pvt. Ltd.*
June 05, 2025 - June 20, 2025 | Control Panel Building, Marine Automation

- Gained hands-on exposure to control panel building and electrical wiring within industrial automation systems.
- Interpreted electrical schematics to support panel assembly and wiring accuracy.
- Assisted with electrical testing procedures for control panels.
- Acquired practical understanding of marine control systems and industrial automation practices.

SELECTED ENGINEERING PROJECTS

AMPET - AI-Based Smart Energy Theft Detection System

- Developed an ESP32-based smart energy monitoring and theft-detection system integrating voltage/current sensors with a Flask web dashboard.

SenSafe - Smart Wearable Women Safety System

- Developed a wearable safety device using ESP32 with embedded firmware written in Embedded C.

SERA - Smart Electronics Research Assistant

- Developed a Flutter-based AI research assistant integrating OpenRouter, NVIDIA AI models and REST APIs.

VENOM - AI-Powered Intelligent Robotic Assistant

- Developed an AI-powered robotic assistant combining Arduino, PCA9685 servo control, OpenCV and AI-based interaction.

Smart Parcel Management System

- Developed a Python-based parcel management application using OpenCV for QR code scanning

SatQuery AI — Vision-Language Assistant for Multimodal Remote Sensing Image Analysis

- An edge AI system for analyzing and classifying satellite imagery using deep learning and NVIDIA Jetson deploymen

RESEARCH PUBLICATION

"Exploring the Synergy of Automation, Machine Learning, and Embedded Systems: Components, Applications, and Future Trends"
International Journal of Progressive Research in Engineering Management and Science (IJPREMS) | January 2025

CERTIFICATIONS & TECHNICAL ACTIVITIES

- German A1 Level Certification
- Industrial Training - Navicom Technology International Pvt. Ltd.
- Industrial Training - SMEC Automation Pvt. Ltd.
- National Level IPR Innovation Challenge 2026
- DataXscape National Level Hackathon finalist 23/1250
- Turtle BOT3 ROS Workshop
- Research Publication Certificate – IJPREMS
- IOT and image processing workshop
- PCB fabrication workshop

ACHIEVEMENTS

- Published research paper, "Exploring the Synergy of Automation, Machine Learning, and Embedded Systems," in IJPREMS (January 2025).
- Completed Marine Electronics Industrial Training at Navicom Technology International Pvt. Ltd., with exposure to navigation and communication equipment associated with INS Vikrant.
- Won the project expo 1st price worth 1000rs

AREAS OF INTEREST

Embedded Systems, Internet of Things (IoT), Artificial Intelligence, Robotics, Edge AI, Marine Automation, Firmware Development, Computer Vision, Smart Energy Systems, Industrial Automation